RAJALAKSHMI ENGINEERING COLLEGE

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Started on	Monday, 31 August 2026, 9:35 PM
State	Finished
Completed on	Monday, 31 August 2026, 9:36 PM
Time taken	1 min 18 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }

        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high);
        }

        return findFirstZero(arr, low, mid - 1);
    }
    return -1;
}

int countZeroes(int arr[], int n) {
    int firstZeroIndex = findFirstZero(arr, 0, n - 1);

    if (firstZeroIndex == -1) {
        return 0;
    }

    return n - firstZeroIndex;
}

int main() {
    int m;
    if (scanf("%d", &m) != 1) return 0;

    int *arr = (int *)malloc(m * sizeof(int));
    if (arr == NULL) return 1;

    for (int i = 0; i < m; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 1;
        }
    }

    printf("%d\n", countZeroes(arr, m));

    free(arr);
    return 0;
}

```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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